

PAUL DESMOND

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PROFESSIONAL SUMMARY

Data Scientist with extensive practical experience in coding, statistical modeling, and machine learning across enterprise environments. Proficient in Python, R, SQL, and Scala with a strong foundation in supervised/unsupervised learning, time-series analysis, and predictive modeling. Skilled at building scalable data science solutions, writing clean code, and translating complex data insights into business actions for cross-functional stakeholders and clients. Fully fluent in English with working proficiency in Portuguese.

TECHNICAL SKILLS

- **Programming Languages:** Python, R, SQL, Scala.
- **Machine Learning & AI:** Supervised/Unsupervised Learning, Deep Learning, Time-Series Analysis, Predictive Modeling, Natural Language Processing (NLP), Computer Vision, Statistical Modeling.
- **ML Frameworks & Libraries:** Scikit-learn, TensorFlow, PyTorch, XGBoost, LightGBM, Pandas, NumPy.
- **Big Data & Engineering:** Spark, Hadoop, AWS, Azure, GCP, Data Warehousing, Data Lake Architecture, ETL Pipelines.
- **MLOps & DevOps:** Docker, Kubernetes, CI/CD, Model Monitoring, A/B Testing, Automated Testing Frameworks.
- **Databases & Tools:** MySQL, PostgreSQL, MongoDB, Cassandra, Tableau, Power BI, Matplotlib, Seaborn.
- **Soft & Professional Skills:** Cross-functional Collaboration, Stakeholder Communication, Analytical Problem Solving, Client Presentations, Strategic Planning.
- **Languages:** English (Native/Fluent), Portuguese (Working Proficiency / Intermediate).

WORK EXPERIENCE

Data Scientist / Data Science Specialist

Danaher Corporation | Cortland, MA | April 2017 – March 2021

- Applied mathematics, statistics, and machine learning to build end-to-end predictive models and ML pipelines for operations and predictive maintenance, reducing equipment downtime by 23%.
- Collaborated with multidisciplinary teams across engineering, marketing, and sales to solve high-impact business problems and deliver scalable data science products.
- Written clean, efficient code in Python, SQL, and Scala to optimize proprietary algorithms and improve baseline model accuracy by 23%.
- Presented complex analytical findings, model performance results, and actionable insights to executive leadership, clients, and cross-functional stakeholders.

Data Scientist

Abiomed | Danvers, MA | September 2014 – February 2017

- Designed statistical features and machine learning algorithms (XGBoost, Random Forest, LSTM) for physiological time-series data, increasing signal detection accuracy by 31%.
- Applied statistical modeling, regression analysis, and experimental design to validate algorithms across multiple test environments.

- Built automated testing frameworks and CI/CD validation pipelines, reducing manual model testing time by 65%.
- Partnered with embedded software engineers and clinical teams to deploy models and communicate technical results effectively.

Data Analyst / Technical Data Scientist

Dell Technologies | Hopkinton, MA | January 2013 – August 2014

- Developed AIOps machine learning models and predictive analytics solutions for enterprise clients, increasing operational efficiency by 35%.
- Processed and analyzed 10+ TB of operational dataset daily using Python, R, and SQL to support anomaly detection and resource optimization.
- Translated client business requirements into technical specs, data pipelines, and actionable recommendations

EDUCATION

Bachelor of Science / Master of Science in Data Science & Mathematics

Bay Path University | Longmeadow, MA | Graduated 2010

- **GPA:** 3.8 / 4.0
- **Coursework:** Advanced Machine Learning, Statistical Inference, Computer Science, Linear Algebra, Data Mining, Time-Series Analysis.

CERTIFICATIONS

- Microsoft Certified: Azure Data Scientist Associate
- AWS Certified Machine Learning – Specialty
- Microsoft Professional Certificate in Data Science